



VIT-AP
UNIVERSITY

Module 2: Cloud-Enabling Technologies

Virtualization Technology

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OUTLINE

- **Cloud-Enabling Technologies**

Broadband Networks and Internet Architecture

foundational network and internet technologies that enable cloud computing, including the role of broadband in providing high-speed data access.

Data Center Technology

Covers the technology and infrastructure used in data centers, which are critical to cloud computing as they house the servers, storage, and networking equipment that deliver cloud services.

Virtualization Technology

that allows multiple virtual machines (VMs) to run on a single physical machine, enabling efficient resource utilization and scalability in cloud environments.

Web Technology

Focuses on the web-based technologies that support cloud computing, such as web servers, APIs, and protocols that facilitate interaction between cloud services and users.

Multitenant Technology

Describes the technology that allows multiple users or tenants to share the same computing resources in a cloud environment while keeping their data and applications isolated from each other.

Containerization

Discusses containerization, a lightweight form of virtualization where applications and their dependencies are packaged together into containers, enabling consistent deployment across different environments.

- **Summary**

- **Virtualization Technology**
 - **Virtualization is the process of converting a physical IT resource into a virtual IT resource.**
 - **Most types of IT resources can be virtualized, including:**
 - **Servers** – A physical server can be abstracted into a virtual server.
 - **Storage** – A physical storage device can be abstracted into a virtual storage device or a virtual disk.
 - **Network** – Physical routers and switches can be abstracted into logical network fabrics, such as VLANs.
 - **Power** – A physical UPS and power distribution units can be abstracted into what are commonly referred to as virtual UPSs.

Virtualization Technology

- ❑ Virtualization Technology
- ❑ Key points:

- **Key points:**
 - The process begins with the allocation of physical IT resources through virtualization software.
 - A guest operating system is then installed on the virtual server.
 - Virtual server and virtual machine (VM) are used synonymously.
 - Each virtual server runs its own guest operating system. The **guest OS operates** independently of the host OS.
 - The guest operating system and application software are unaware of the virtualization process.
 - Programs and applications run on virtual servers as if they were on separate physical servers.
 - Virtualized IT resources are executed as if they were running on physical systems.
 - Software products and applications typically do not require customization, configuration, or patching to run in a virtualized environment.
 - The physical server on which virtualization software runs is called the **host or physical host**.
 - Virtualization Software (Hypervisor):
 - Provides system services related to virtual machine management. Often referred to as a virtual machine manager (VMM) or hypervisor.

Virtualization Technology

- ❑ Virtualization Technology
- ❑ Key points:
- ❑ Hardware Independence

- Hardware Independence

- Hardware Dependencies in Non-Virtualized Environments:

- Operating systems are configured for specific hardware models.
 - Reconfiguration is needed if IT resources are modified.

- Virtualization and Hardware Independence:

- Virtualization converts unique IT hardware into emulated, standardized software-based copies.
 - Virtual servers can easily be moved to another virtualization host.

- Benefits of Hardware Independence:

- Automatically resolves hardware-software incompatibility issues.
 - Simplifies the cloning and manipulation of virtual IT resources compared to physical hardware.

Virtualization Technology

- ❑ Virtualization Technology
- ❑ Key points:
- ❑ Hardware Independence
- ❑ Server Consolidation

- **Server Consolidation**

- Virtualization software allows multiple virtual servers to be created simultaneously on the same host.
- Different virtual servers share one physical server through server consolidation.
- Benefits of Server Consolidation:
 - Increases hardware utilization.
 - Supports load balancing.
 - Optimizes available IT resources.
 - Flexibility of Virtual Servers:
 - Enables different virtual servers to run different guest operating systems on the same host.
 - Support for Cloud Features:
 - Server consolidation underpins key cloud features like on-demand usage, resource pooling, elasticity, scalability, and resiliency.

Virtualization Technology

- ❑ Virtualization Technology
- ❑ Key points:
- ❑ Hardware Independence
- ❑ Server Consolidation
- ❑ Resource Replication

- Resource Replication

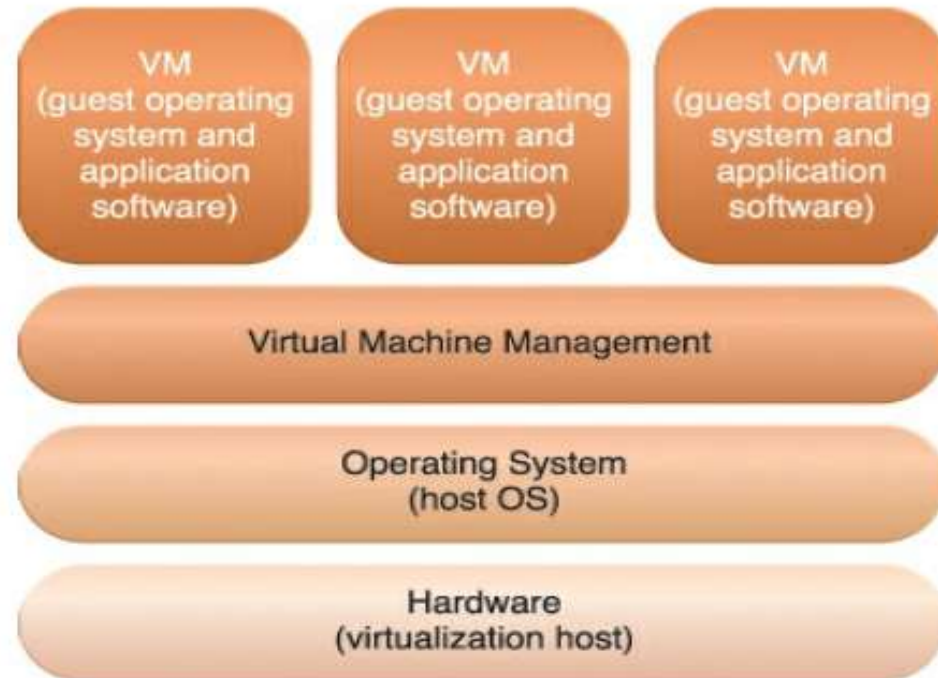
- Virtual servers are created as virtual disk images containing binary file copies of hard disk content.
- These virtual disk images are accessible to the host's operating system, enabling easy file operations.
- Ease of Manipulation and Replication:
 - Simple file operations (copy, move, paste) can be used to replicate, migrate, and back up virtual servers.
- Key Benefits of Resource Replication:
 - Allows pre-packaging of virtual disk images with virtual hardware capabilities, guest operating systems, and application software for instant deployment.
 - Facilitates rapid scaling out and up by quickly migrating and deploying new instances of virtual machines.
 - Instantaneous creation of VM snapshots by saving the state of the virtual server's memory and hard disk image to a host-based file.
 - Operators can easily revert to these snapshots to restore the virtual machine to a previous state.
 - Enables efficient backup and restoration procedures.
 - Allows the creation of multiple instances of critical IT resources and applications for redundancy and disaster recovery.

Virtualization Technology

- ❑ Virtualization Technology
- ❑ Key points:
- ❑ Hardware Independence
- ❑ Server Consolidation
- ❑ Resource Replication
- ❑ Operating System-Based Virtualization

- Operating System-Based Virtualization

- Operating system-based virtualization involves installing virtualization software on an existing operating system, known as the **host operating system**.
- Runs multiple virtual machines (VMs) on a single OS kernel, sharing the same operating system and resources, which makes it more lightweight and efficient in resource usage.



Virtualization Technology

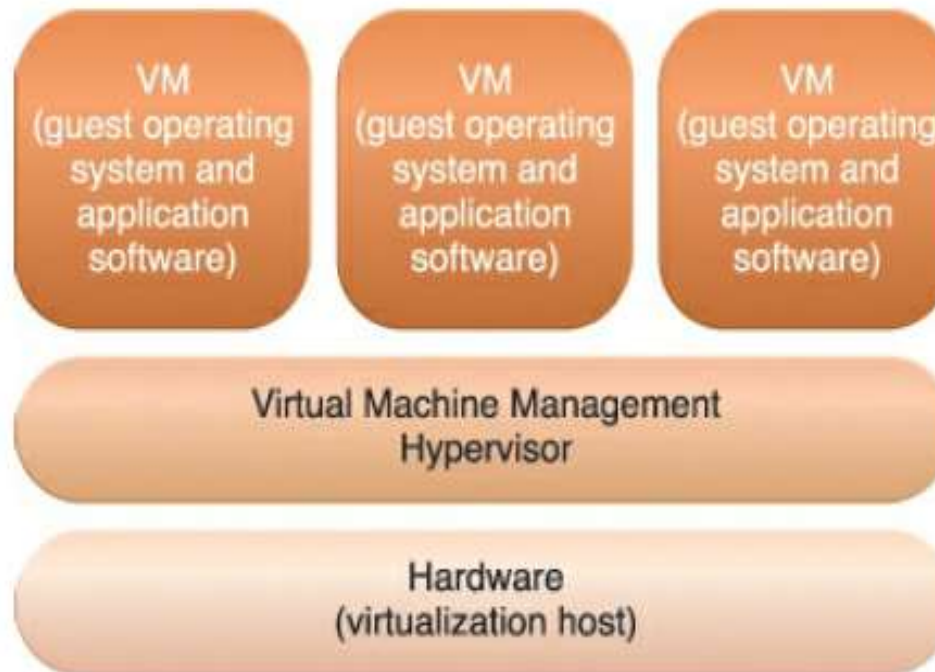
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- ❑ Server Consolidation
- ❑ Resource Replication
- ❑ Operating System-Based Virtualization

- The host operating system provides hardware support,
 - which can resolve hardware compatibility issues even if the virtualization software lacks specific hardware drivers.
- operating **system-based services** that are available as administration tools can be used to manage the physical host.
 - • Backup and Recovery
 - • Integration to Directory Services
 - • Security Management
- Operating system-based virtualization can introduce **demands and issues** related to performance overhead
 - The host operating system consumes CPU, memory, and other hardware IT resources.
 - Hardware-related calls from guest operating systems need to traverse several layers to and from the hardware, which decreases overall performance.
 - Licenses are usually required for host operating systems, in addition to individual licenses for each of their guest operating systems.

Virtualization Technology

- ❑ Virtualization Technology
- ❑ Key points:
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- ❑ Server Consolidation
- ❑ Resource Replication
- ❑ Operating System-Based Virtualization
- ❑ Hardware-Based Virtualization

- **Hardware-Based Virtualization**
 - the installation of virtualization software directly on the physical host hardware so as to
 - bypass the host operating system
- Utilizes hardware features to create and run multiple VMs, each with its own OS and complete hardware abstraction, offering better isolation and compatibility but generally with more overhead.



Virtualization Technology

Hypervisor:

- Virtualization software is commonly referred to as a hypervisor.
- A hypervisor is a thin software layer that manages hardware functions to establish a virtualization management layer.
- The hypervisor has a simple user interface and requires minimal storage space.
- Device drivers and system services within the hypervisor are optimized for provisioning virtual servers.
- Many standard operating system functions are not implemented to reduce performance overhead and optimize the management of multiple virtual servers on the same hardware.

Compatibility Issues:

Hardware Device Compatibility:

Device drivers and support software must be compatible with the hypervisor.

Host Management and Administration:

Hypervisors may lack the advanced management and administration features commonly found in standard operating systems.

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- ❑ Server Consolidation
- ❑ Resource Replication
- ❑ Operating System-Based Virtualization
- ❑ Hardware-Based Virtualization
- ❑ Hypervisor:

Virtualization Technology

- Virtualization Management

- Ease of Administration: Virtual servers simplify administrative tasks compared to physical servers.
- Modern virtualization software includes advanced management functions that automate tasks and reduce the operational burden on IT resources.
- Virtualization Infrastructure Management (VIM)
 - VIM tools manage virtual IT resources collectively.
- A centralized management module, or controller, typically runs on a dedicated computer to oversee virtual resources.

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- ❑ Operating System-Based Virtualization
- ❑ Hardware-Based Virtualization
- ❑ Hypervisor:
- ❑ Virtualization Management

Virtualization Technology

- Other Considerations

- Performance Overhead: Virtualization may not be suitable for complex systems with high workloads that do not benefit from resource sharing or replication.
- Para-Virtualization: A technique to reduce performance overhead by presenting a modified software interface to virtual machines, rather than changing the underlying hardware.
- Special Hardware Compatibility: Compatibility issues can arise if specialized hardware lacks device drivers compatible with virtualization software, or if the software is incompatible with newer hardware versions.

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- ❑ Hardware-Based Virtualization
- ❑ Hypervisor:
- ❑ Virtualization Management
- ❑ Other Considerations

- ☐ **Virtualization Technology**
- ☐ **Key points**
- ☐ **Hardware Independence**
- ☐ **Server Consolidation**
- ☐ **Resource Replication**
- ☐ **Operating System-Based Virtualization**
- ☐ **Hardware-Based Virtualization**
- ☐ **Virtualization Management**
- ☐ **Other Considerations**

References

1. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, “Cloud Computing: Concepts, Technology & Architecture”, PHI Publications, 2013.

Quiz

**Thank
You**